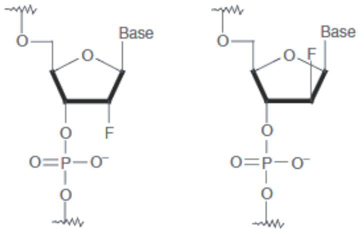
2′-Fluoro & 2′-Fluoro-arabino



2'-fluoro DNA (FNA)

2'-deoxy-2'-fluoro-D-arabino nucleic acid (FANA)

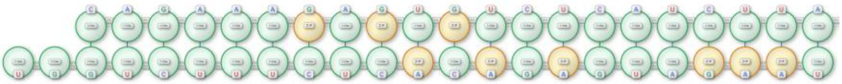
2′-Fluoro modifications, including 2′F-ANA (fluoro-arabino nucleic acid) and related FNA chemistries are highly potent nucleic acid modifications in which a fluorine atom replaces the 2′-hydroxyl group of the sugar. This change enhances nuclease stability and strengthens target binding, while also supporting steric-blocking mechanisms; in some cases, these modifications can enable gymnotic cellular uptake without delivery systems.

Inclisiran (Leqvio)

Inclisiran (Leqvio) is a chemically modified, double-stranded small interfering RNA (siRNA) conjugated to GalNAc to tissue-selectively knock down PCSK9 mRNA, leading to reduced hepatic PCSK9 synthesis and increased LDL receptor recycling, thereby lowering circulating low-density lipoprotein cholesterol (LDL-C) (FDA, 2021; EMA, 2020) . It is indicated as an adjunct to diet and lipid-lowering therapy, including statins, for patients with primary hypercholesterolemia or atherosclerotic cardiovascular disease who require additional LDL-C reduction, and is administered subcutaneously with an extended dosing interval of every 6 months; illustrating a robust and long hepatic half-life (FDA, 2021; EMA, 2020).

Nonclinical and clinical studies demonstrate dose-dependent and sustained reductions in PCSK9 and LDL-C, with distribution primarily to the liver, metabolism via nuclease degradation to shorter RNA fragments, and elimination mainly through urine and feces (FDA, 2021). Toxicological findings were largely non-adverse and associated with tissue accumulation, with no evidence of genotoxicity, carcinogenicity, or reproductive toxicity, supporting a favorable nonclinical safety profile (FDA, 2021).

Givosiran (GIVLAARI)



Givosiran (GIVLAARI) is a small interfering RNA (siRNA) therapeutic indicated for the treatment of adults with acute hepatic porphyria (AHP). It acts by targeting Aminolevulinate Synthase 1 (ALAS1) mRNA in hepatocytes, thereby reducing the accumulation of neurotoxic intermediates such as aminolaevulinic acid (ALA) and porphobilinogen (PBG). The drug is administered once a month through subcutaneous injection at a dose of 2.5 mg/kg (Alnylam Pharmaceuticals, Inc., 2019) or about 180mg per dose for an average person.

Clinical studies have demonstrated that Givosiran significantly reduces the frequency of porphyria attacks, with patients experiencing approximately 70% fewer attacks compared to placebo in the ENVISION trial *(page 10 on the drug label/US Prescribing Information (USPI)*. Common adverse effects include nausea and injection-site reactions, while more serious risks include hepatic toxicity, renal impairment, and rare anaphylactic reactions (Alnylam Pharmaceuticals, Inc., 2019).

2′F-ANA (2′-fluoro-arabino nucleic acid) is a chemically modified nucleotide in which substitution at the 2′ position of the sugar enhances nuclease resistance and increases binding affinity to complementary RNA, while maintaining compatibility with RNase H activity in gapmer designs (Souleimanian et al., 2012) . These properties have made 2′F-ANA a useful modification in preclinical studies, where it has demonstrated potent and sustained gene silencing. However, despite its favorable biochemical profile, this chemistry has not yet been translated into clinically approved oligonucleotide therapeutics and remains primarily confined to experimental applications.